

•



- [Core Concepts](#)
- [Models](#)
- White-Box Explanations

On this page

## White-Box Explanations

Was this page helpful? 

White-box explanations (WBE) is a tool that aims at providing explanations for predictions of machine learning models in Case Manager (CM). It reduces the amount of time data scientists spend setting up rules that will generate explanations in CM (i.e. less than 2 hours) and allows analysts in CM to understand why a transaction received a certain score, including a breakdown of the contributions to that score.

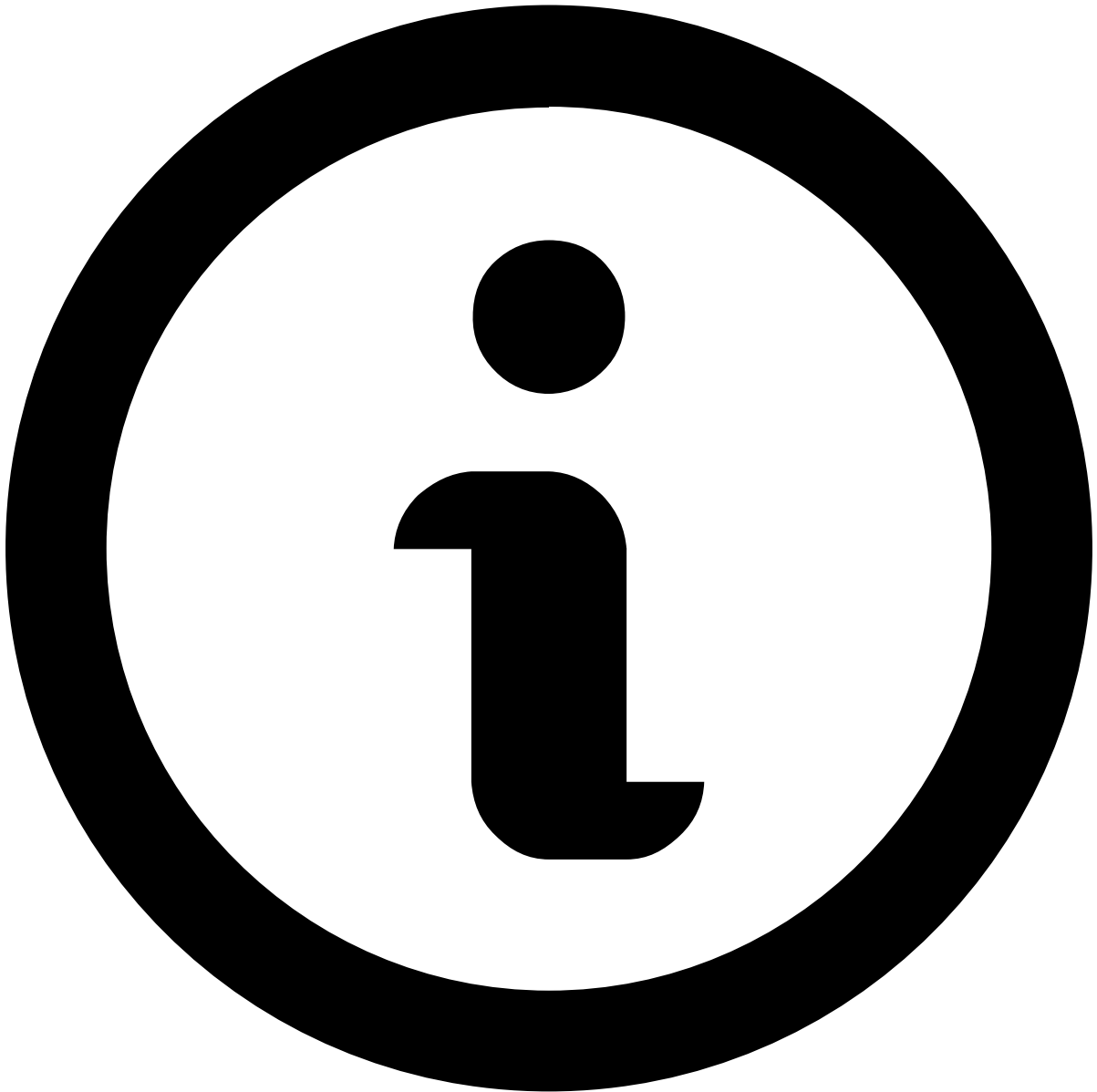
As a result of providing explanations for each transaction score in CM, WBE supports analysts in two main aspects:

- Guidance on how to proceed when they need to understand the reason behind a transaction alert.
- Insight on why the model is delivering high scores to certain transactions.

## Introduction to WBE

WBE lists the most important features that influence a transaction's score in CM, allowing analysts to understand the reason behind that score.

The flow of WBE starts in the model, where you can configure explanations. During this configuration, data scientists can add explanation messages. After that, the model is put into Production (i.e. workflow), where it will score and create explanations for transactions. Finally, analysts using CM will be able to see the explanation messages previously generated by Pulse at the end of the workflow.



Explanations in model evaluation

Currently it is not possible to generate explanations in the model evaluation.

## WBE algorithm␣

Depending on the algorithm you use to train the model, the following algorithms can be used for the explanations:

- **Feedzai Feature Contribution:** an algorithm based on the fraud ratio gain, for models trained with the [Feedzai Random Forest](#) algorithm.
- **TreeSHAP Feature Contribution:** an algorithm based on [Shapley values](#), for models trained with Microsoft LightGBM or Feedzai FairGBM.



### Note

Explanations provided through Shapley values, which are our Feature Contribution (FC) weights, are computed based on the prevalence ratio of the dataset (e.g., the rate of fraudulent transactions) ␣the base value. Each FC weight demonstrates how much that feature contributed to the difference between the prediction score and that base value. Therefore, if the prediction score is higher than the base value, the sum of the normalized FC weights must be 100% of the prediction score. However, if the prediction score is lower than the base value, the sum of the normalized FC weights will be -100%. This is a rare event in financial crime use cases, given the low rate of financial crime when compared to the whole set of transactions.

## Learn more

Check the following pages in the User Guide to learn how to put WBE into practice:

- To set up WBE, go to the [Training a Model](#) page and read the Training Parameters section.
- To check how WBE are produced in runtime, read the [Adding Models](#) page.